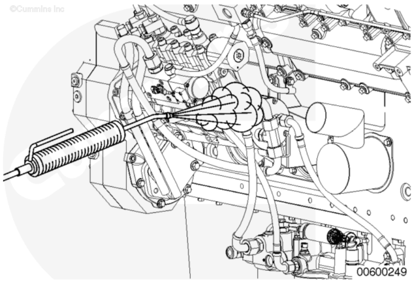
## Trainings

## Preparatory Steps




**WARNING**

**When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.**

**WARNING**

**Some solvents are flammable and toxic. Read the manufacturer's instructions before use.**

**WARNING**

**Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.**

**WARNING**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**

Disconnect the batteries. See equipment manufacturer service information.

Thoroughly clean debris from all fittings and components before removal. Make sure that debris, water, steam, or cleaning solution do **not** get inside the fuel system.

Use compressed air to dry the components.

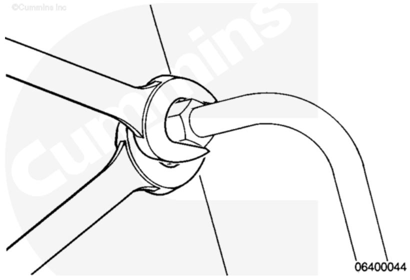
# Remove

## with Mechanically Actuated Injector

Use two wrenches to remove the fuel tubes and hoses.

Support the mating fittings with a wrench. Loosen the fuel tube nuts with the other wrench.

Remove the fuel tubes and hoses.

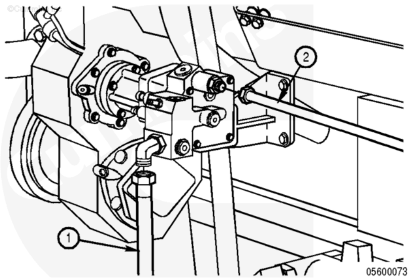



**⚠ WARNING ⚠**

**Depending on the circumstances, fuel is flammable. When performing any or all of the following procedures to remove supply lines and related components, keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on a fuel system.**

**⚠ CAUTION ⚠**

**Very small amounts of dirt and debris can be very harmful to the injectors and the cone seats on the injectors. Extra care is required to keep the connections clean during removal and installation. The fuel supply line(s) must be placed in a plastic bag or connections be covered immediately to keep them clean when removed from the engine. Failure to do so can cause fuel system failure.**




Disconnect the fuel inlet (1).

Disconnect the electronic fuel control valve supply tube (2).

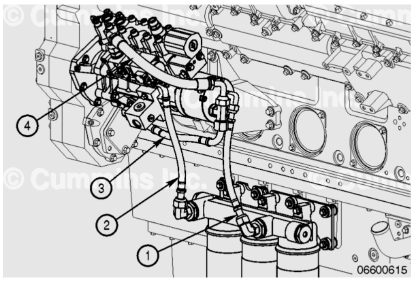
Remove the o-rings.

## with Electronically Actuated Injector

Remove the following items and cover the fittings with a plastic bag, or place the items inside a plastic bag.

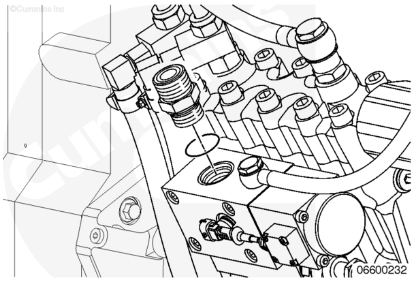
1. Outlet of the gerotor to inlet of stage two filter

2. Outlet of stage two filter to the inlet of the fuel pump pressurizing assembly
3. Fuel supply to gerotor
4. Fuel bypass to gerotor.



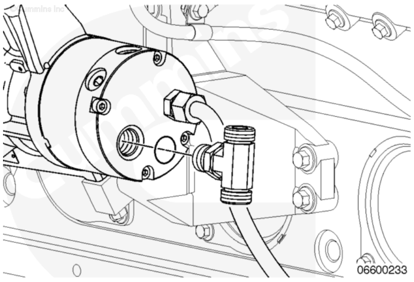

Remove the following and cover the fittings with a plastic bag, or place the items inside a plastic bag.

Remove the union from the fuel pump and discard the o-ring.



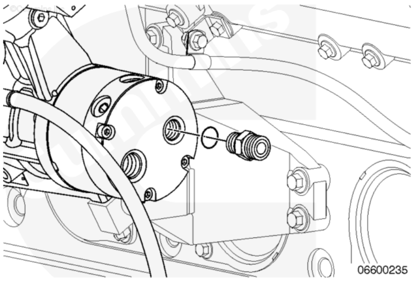

Remove the following and cover the fittings with a plastic bag, or place the items inside a plastic bag.

Remove the tee fitting from the gerotor fuel pump. Discard the o-rings.



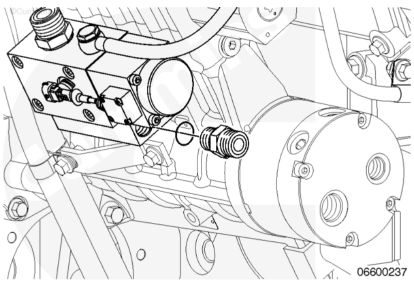

Remove the following and cover the fittings with a plastic bag, or place the items inside a plastic bag.

Remove the union from the gerotor pump. Discard the o-rings.




Remove the following and cover the fittings with a plastic bag, or place the items inside a plastic bag.

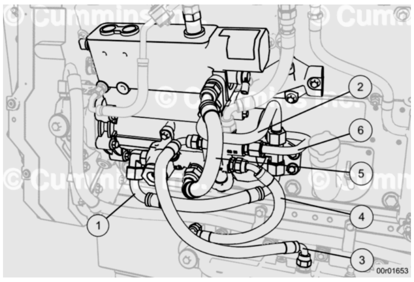
Remove the union from the inlet of the fuel pump. Discard the o-ring.




**For Generator Model DQKAN with Low NOx Emissions Capability:**

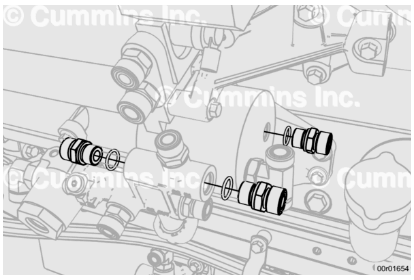
Remove the following items and cover the fittings with a plastic bag, or place the items inside a plastic bag.

1. Fuel supply to gerotor
2. Fuel bypass to gerotor
3. Stage 2 filter inlet tube
4. Stage 2 filter outlet tube
5. Pump Inlet supply hose
6. Gerotor outlet tube



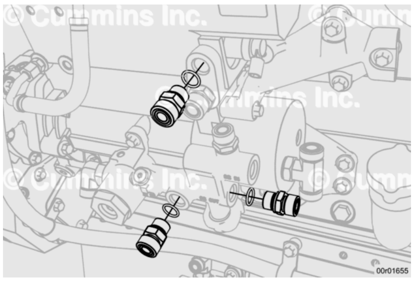
Remove the unions from the fuel inlet manifold and fuel pump gerotor, and cover with a plastic bag, or place the item in a plastic bag.

Discard o-rings.



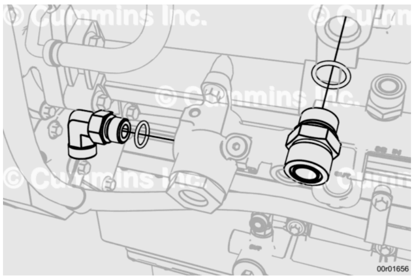
Remove the unions from the fuel inlet manifold and fuel pump, and cover with a plastic bag, or place the item in a plastic bag.

Discard o-rings.



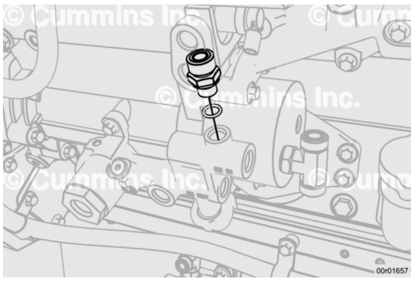
Remove the union and tee fitting from the fuel inlet manifold and fuel pump, and cover with a plastic bag, or place the item in a plastic bag.

Discard o-rings.



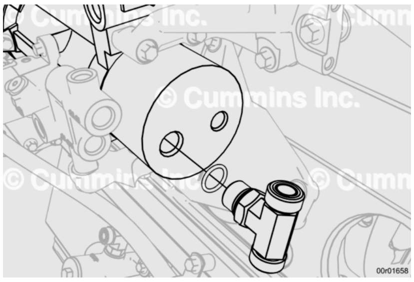
Remove the union from the fuel inlet manifold and cover with a plastic bag, or place the item in a plastic bag.

Discard o-rings.



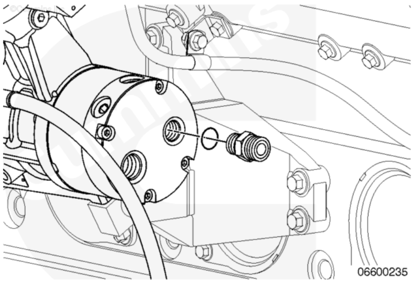
Remove the tee fitting from the gerotor fuel pump the and cover the fitting with a plastic bag, or place the item inside a plastic bag.

Discard the o-rings.




Remove the union from the gerotor and cover the fittings with a plastic bag, or place the items inside a plastic bag.

Remove the union from the gerotor pump. Discard the o-rings.

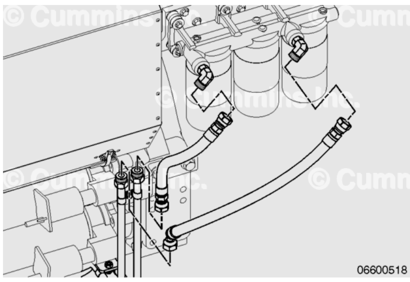



## Marine Applications

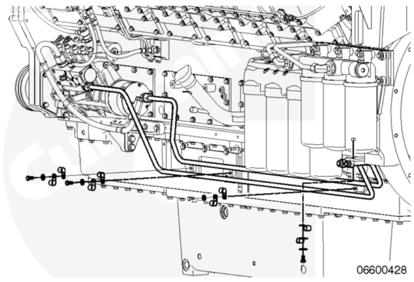
Remove the following and cover the fittings with a plastic bag, or place the items in a plastic bag.

Remove the fuel supply lines to the inlet and outlet of the stage two filter head.

Discard the o-rings.




Remove the capscrews and clamps securing the fuel line to the mounting plates.

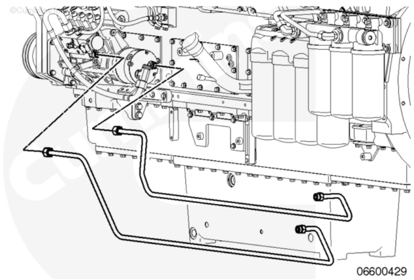



Remove the following and cover the fittings with a plastic bag, or place the items in a plastic bag.

Disconnect the fuel supply lines at the bottom of the pan rail.

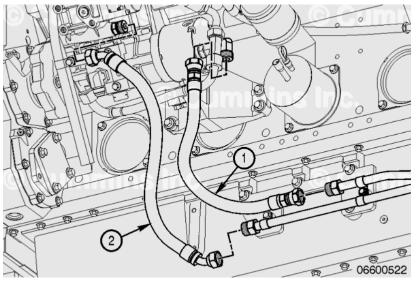
Remove the fuel supply lines to the remote mounted stage 2 filter.

Discard the o-rings.

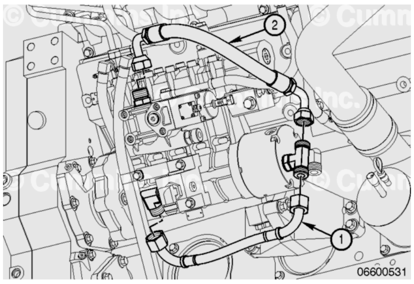


Remove the following and cover the fittings with a plastic bag, or place the items in a plastic bag.

1. Outlet of the gerotor to inlet of stage two filter.
2. Outlet of stage two filter to the inlet of the fuel pump pressurizing assembly.

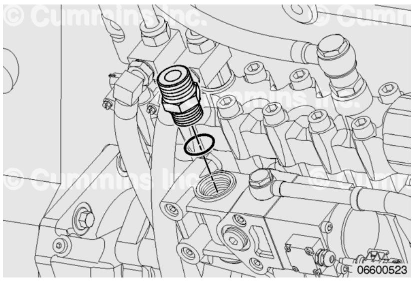



1. Fuel supply to gerotor.
2. Fuel bypass to gerotor.




Remove the following and cover the fittings with a plastic bag, or place the items inside a plastic bag.

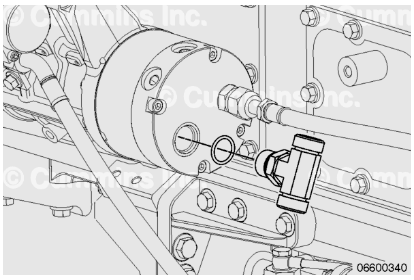
Remove the union from the fuel pump pressurizing assembly and discard the o-ring.




Remove the following and cover the fittings with a plastic bag, or place the items inside a plastic bag.

Remove the tee fitting from the gerotor fuel pump.

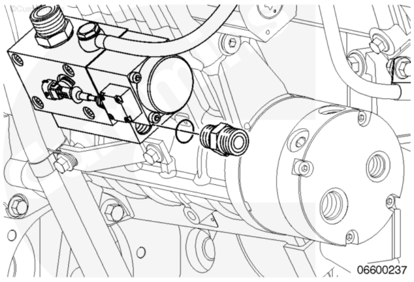
Discard the o-rings.




Remove the following and cover the fittings with a plastic bag, or place the items in a plastic bag.

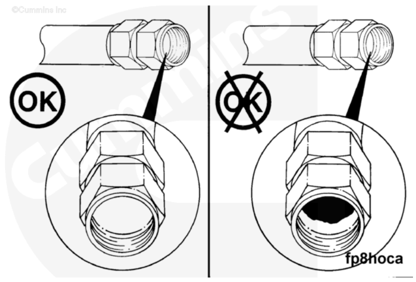
Remove the union from the inlet of the fuel pump actuator.

Discard the o-rings.



# Clean and Inspect for Reuse

## with Mechanically Actuated Injector




Inspect the inside of the hose.

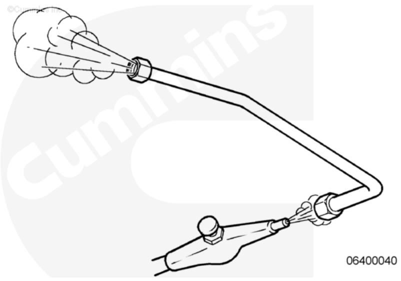
- The inner lining of the hose can separate from the center hose section.
- A separation of the inner lining can cause a restriction in the fuel flow.

Inspect for any pinches in the hose that can obstruct the flow.

The hose **must not** have pinches or loops that can obstruct the flow.

**⚠ WARNING ⚠**

**Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.**

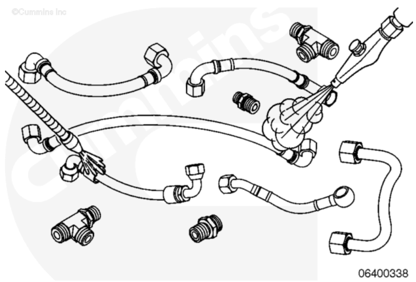



Use compressed air to flush the lines to remove any loose dirt particles.

## with Electronically Actuated Injector

**⚠ WARNING ⚠**

**When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.**




**⚠ WARNING ⚠**

**Some solvents are flammable and toxic. Read the manufacturer's instructions before use.**

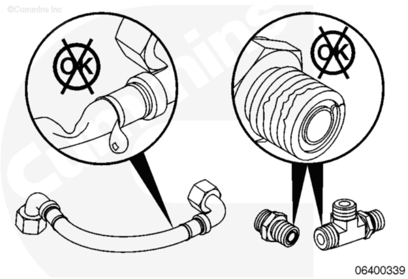
**⚠ WARNING ⚠**

**Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.**

Use solvent to clean the fuel supply lines, tee fittings, and unions.

Dry with compressed air.

Inspect the fuel supply lines for restrictions that can possibly prevent fuel from flowing freely. Replace the fuel supply lines if kinks, leakage, or other damage is found.



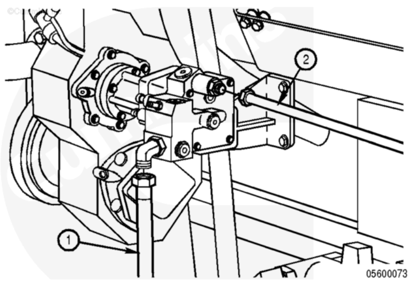

Inspect the tee fittings and unions for damage. Replace if cracked, otherwise damaged, or if stripped threads are found.

Cover the fuel line fittings with a plastic bag, or place the items inside a plastic bag after cleaning and inspection.

# Install

## with Mechanically Actuated Injector

Install new o-rings in the fuel pump inlet hose fitting and the fuel supply tube fitting.




Install the fuel pump inlet hose (1) and the fuel supply tube (2).

Tighten the hoses.

**Torque Value:**
Fuel Pump Inlet Hose

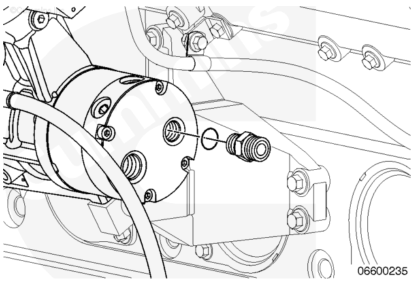
1. 120 n•m [ 89 ft-lb ]

**Torque Value:**
Fuel Supply Tube

1. 120 n•m [ 89 ft-lb ]

## with Electronically Actuated Injector

Remove the union from its protective plastic bag and install new o-rings on the union.



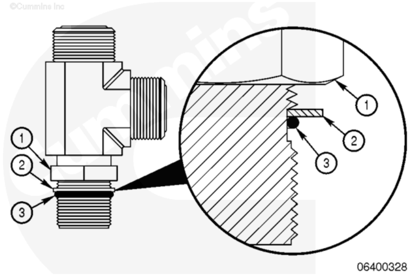

Install the union in the outlet of the gerotor fuel pump.

**Torque Value:**
Gerotor Pump Outlet Union

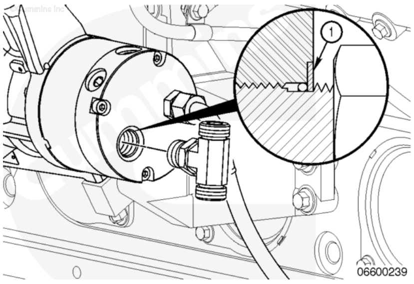
1. 100 n•m [ 74 ft-lb ]

Remove the tee fitting from its protective plastic bag and install the tee fitting in the port on the back of the fuel pump by first positioning the locknut.

1. Locknut
2. Washer
3. O-ring.




Install the tee fitting in the fuel pump port until the backup washer (1) contacts the port face.

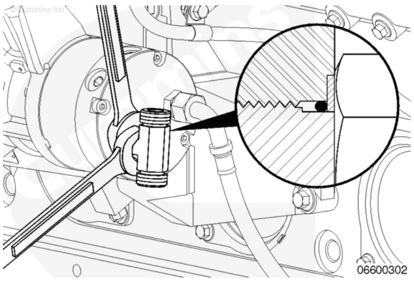



Using two wrenches, hold the tee fitting in the desired position and tighten the locknut.

**Torque Value:**

Tee Fitting Locknut

1. 100 n•m [ 74 ft-lb ]



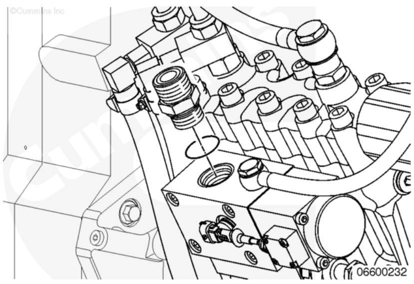

Remove the union from its protective plastic bag

Install new o-rings in the union, and then install it in the port for the bypass from the fuel pump inlet actuator.

**Torque Value:**

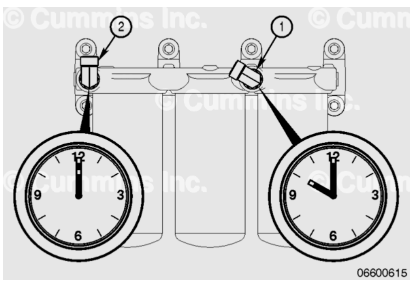
Fuel Pump Inlet to Actuator Union

1. 170 n•m [ 125 ft-lb ]




Make sure the fuel inlet (1) and outlet (2) elbow joints are positioned at a 10 o'clock and 12 o'clock angle as shown.

If elbow joints need to be repositioned, see the assembly steps in Fuel Filter Head. Refer to Procedure 006-017 in section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-017-tr.html)




Remove the fuel line from its protective plastic bag. Install the fuel line from the outlet of the stage two filters to the inlet of the fuel pump pressurizing assembly.

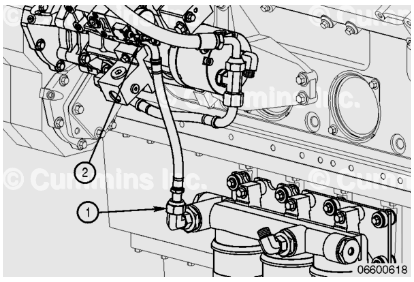
Tighten the hose fitting (1) to the elbow of the filter head.

Tighten the hose fitting at the union (2) on the fuel pump pressurizing assembly.

Hose fitting sizes differ depending on hose part number.

**Torque Value:** 65 n•m [ 48 ft-lb ]

**Torque Value:** 90 n•m [ 66 ft-lb ]




Remove the fuel line from its protective plastic bag. Install the fuel supply line from the gerotor fuel pump to the inlet of the stage two fuel filter.

Tighten the hose fitting (1) to the elbow of the filter head.

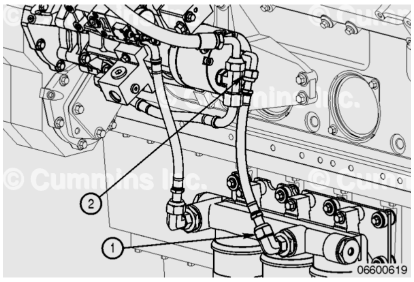
Tighten the hose fitting at the union (2) on the gerotor fuel pump.

Hose fitting sizes differ depending on hose part number.

**Torque Value:** 65 n•m [ 48 ft-lb ]

**Torque Value:** 90 n•m [ 66 ft-lb ]

**Note : Make sure that both elbow unions are still positioned at the proper angles as described above. Avoid sharp bends or twisting of the hose. Some small adjustment to the elbow fittings may be required to reduce the risk of applying stress to the hose.**



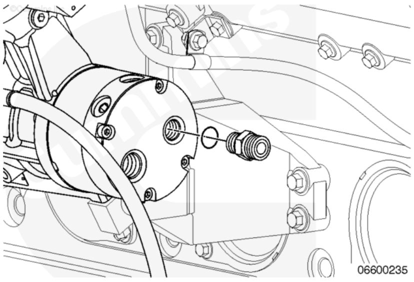

**For Generator Model DQKAN with Low NOx Emissions Capability:**

Remove the union from its protective plastic bag and install new o-rings.

Install the union to the outlet of the gerotor fuel pump.

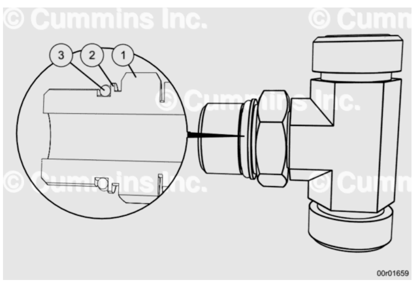
**Torque Value:**
Gerotor Pump Outlet Union

1. 85 n•m [ 63 ft-lb ]



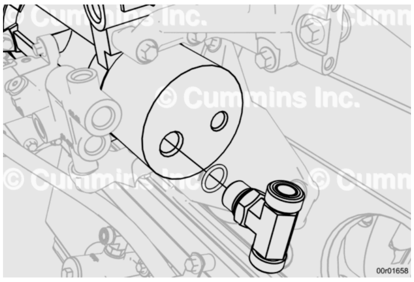

Remove the tee fitting from its protective plastic bag and install the tee fitting in the port on the back of the fuel pump by first positioning the locknut.

1. Locknut
2. Washer
3. O-ring.



Install the tee fitting in the fuel pump port until the backup washer (1) contacts the port face.

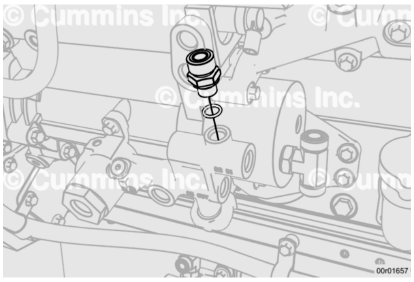
Do **not** torque.



Remove the union from its protective plastic bag and install to fuel inlet manifold.

Install new o-rings.

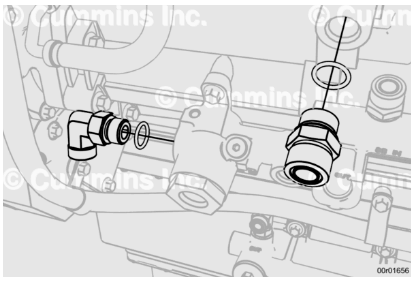
**Torque Value:** 45 n•m [ 33 ft-lb ]



Remove the union and elbow from their protective plastic bags and install to the fuel manifold and fuel pump.

Install new o-rings.

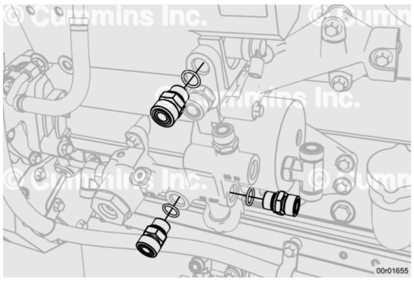
**Torque Value:** 140 n•m [ 103 ft-lb ]



Remove the unions from their protective plastic bags and install to the fuel inlet manifold and fuel pump.

Install new o-rings.

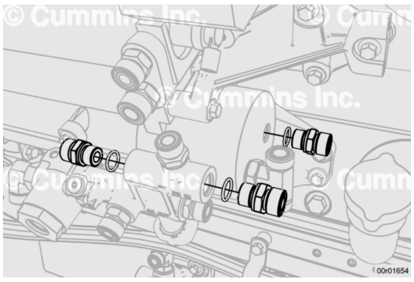
**Torque Value:** 85 n•m [ 63 ft-lb ]



Remove the unions from their protective plastic bags and install to the fuel inlet manifold and fuel pump gerotor.

Install new o-rings.

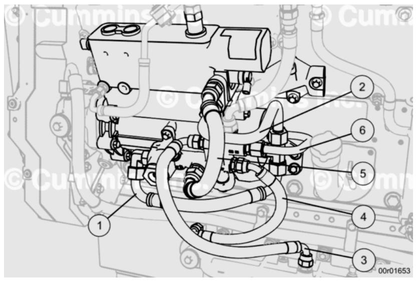
**Torque Value:** 85 n•m [ 63 ft-lb ]



Remove the following items from plastic bags and install.

1. Fuel supply to gerotor
2. Fuel bypass to gerotor

3. Stage 2 filter inlet tube
4. Stage 2 filter outlet tube
5. Pump Inlet supply hose
6. Gerotor outlet tube



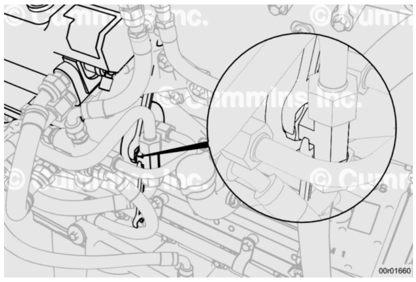

### Torque Value

1. Fuel supply to gerotor: 115 n•m [85 ft-lb ]
2. Fuel bypass to gerotor: 115 n•m [85 ft-lb ]
3. Stage 2 filter inlet tube: 85 n•m [63 ft-lb ]
4. Stage 2 filter outlet tube: 85 n•m [63 ft-lb ]
5. Pump Inlet supply hose: 115 n•m [85 ft-lb ]
6. Gerotor outlet tube: 80 n•m [59 ft-lb ]

**Note : Make sure that both elbow unions are still positioned at the proper angles as described above. Avoid sharp bends or twisting of the hose. Some small adjustment to the elbow fittings may be required to reduce the risk of applying stress to the hose.**

Whilst holding the tee fitting in the position which ensures the best seating with the lines above and below, using two wrenches, tighten the tee fitting locknut.

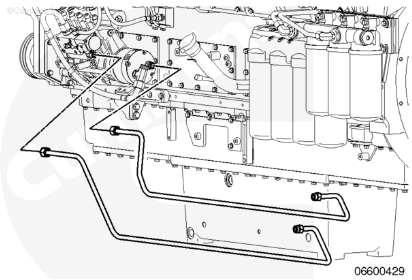
**Torque Value:** 115 n•m [ 85 ft-lb ]




## Marine Applications

Remove the fuel supply lines from the protective plastic bag.

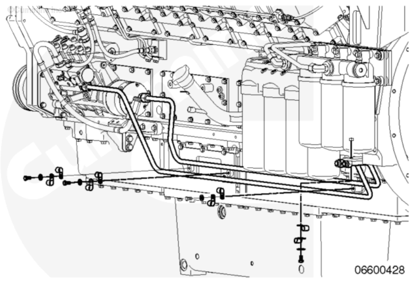
Install the fuel supply lines that go along the lubricating oil pan rail.




Install the clamps and capscrews securing the fuel line to the mounting plates.

Tighten the capscrews.

**Torque Value:** 9 n•m [ 80 in-lb ]

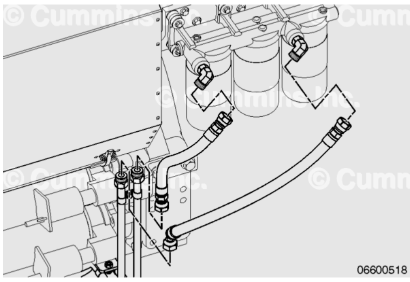


Remove the fuel supply lines from the protective plastic bag.

Install the short section of the fuel supply lines to the inlet and outlet of the stage 2 filter head.

Tighten the fittings.

**Torque Value:** 91 n•m [ 67 ft-lb ]



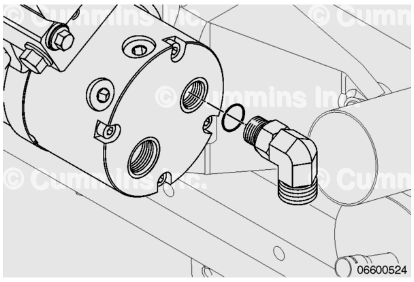

Remove the elbow from its protective plastic bag.

Install new o-rings on the union.

Install the union in the outlet of the gerotor fuel pump.

**Torque Value:**
Gerotor Pump Outlet Union

1. 100 n•m [ 74 ft-lb ]



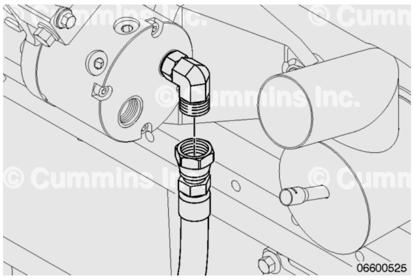

Remove the fuel line from its protective plastic bag.

Install the fuel supply line from the gerotor fuel pump to the tube leading to the inlet of the second stage filter.

Install new o-rings on each end.

**Torque Value:**
Fuel Supply Line

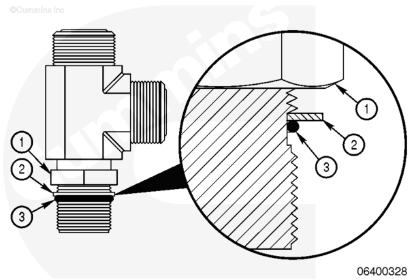
1. 65 n•m [ 48 ft-lb ]



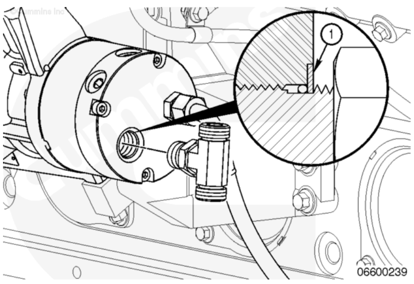

Remove the tee fitting from its protective plastic bag.

Install the tee fitting in the port on the back of the fuel pump by first positioning the locknut.

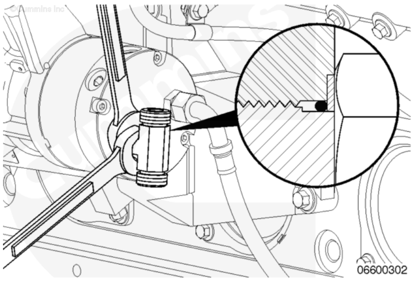
1. Locknut
2. Washer
3. O-ring.




Install the tee fitting in the fuel pump port until the backup washer (1) contacts the port face.



Use two wrenches to hold the tee fitting in the desired position and tighten the locknut.

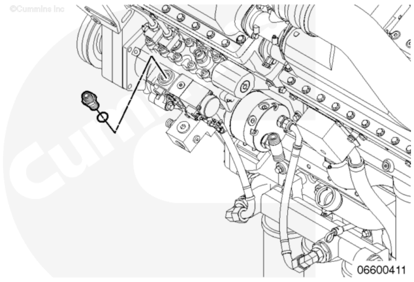



**Torque Value:**
Tee Fitting Locknut
1. 100 n•m [ 74 ft-lb ]

Remove the union from its protective plastic bag.

Install new o-rings in the union.

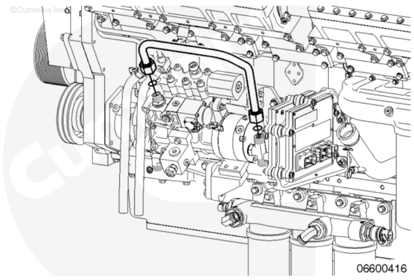
Install the union in the port for the bypass from the fuel pump inlet actuator.




**Torque Value:**
Fuel Pump Inlet to Actuator Union
1. 170 n•m [ 125 ft-lb ]

Remove the fuel supply line from its protective plastic bag.

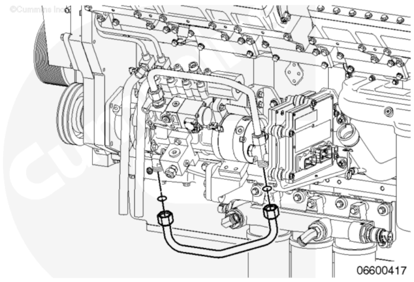
Install the fuel bypass line on the inlet actuator bypass union to the tee fitting on the inlet of the gerotor fuel pump.




**Torque Value:**
Gerotor Fuel Supply Line
1. 91 n•m [ 67 ft-lb ]

Remove the fuel supply line from its protective plastic bag.

Install the fuel supply line from the fuel manifold to the tee fitting on the gerotor fuel pump.



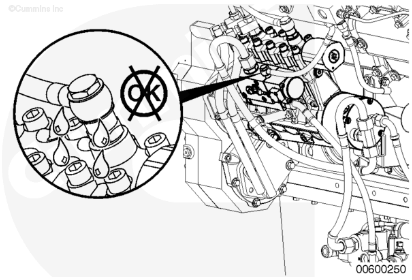

**Torque Value:**
Fuel Line From Manifold to Gerotor Pump Tee Fitting
1. 91 n•m [ 67 ft-lb ]

## Finishing Steps

**⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing,**

**remove the negative (-) battery cable first and attach the negative (-) battery cable last.**




Connect the batteries. See equipment manufacturer service information.

Operate the engine and check for leaks. If leaks are found, repair and test again.

**Last Modified: 17-Aug-2021**